# KUSH PATEL

**AML / Financial Crime Analyst | Transaction Monitoring · Investigations · FINTRAC Reporting · Controls | SQL · Python · Power BI**

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Open to remote and relocation across Canada · Legally eligible to work in Canada · Available immediately

## SUMMARY

- Analyst with 4+ years in finance and controls: month-end GL/P&L reconciliations and risk reporting at an investment firm, then reconciliation controls, exception logging and access controls as sole analyst at a distributor, cutting recurring discrepancies an estimated ~45%.
- Built a transaction-monitoring system on synthetic data that finds all 60 planted cases where a flat $10,000 rule finds 38, while alerting on a third fewer entities, with each rule mapped to FINTRAC's indicator categories and reporting obligations.
- Writes for the reviewer: an analytic brief on a structuring case with a bottom line, key judgments with confidence levels, alternatives considered, a reporting position and the facts still needed before a decision.

## SKILLS

**Financial Crime:** structuring and smurfing typologies, rule design and tuning, alert triage and prioritisation, investigation case files, analytic briefs, PCMLTFA and FINTRAC reporting (LCTR including the 24-hour rule, EFTR, STR timing), threshold versus suspicion

**Detection Analytics:** SQL window functions for velocity and aggregation rules, entity-relative z-scores, anomaly detection (IsolationForest, Mahalanobis distance, gradient boosting), precision and recall by typology, cost curves for alert volume

**Controls & Audit:** reconciliation and control totals, journal-entry testing, segregation of duties, exception logs, evidence integrity, RLS/RBAC and column masking

**Tools:** SQL (T-SQL, SQLite), Python (pandas, scikit-learn), Power BI, Excel, Streamlit, Tableau, Git

## PROFESSIONAL EXPERIENCE

### Two Rivers Specialty Meats
Dec 2023 – Jun 2026 · North Vancouver, BC

**Data Analyst — Operations, Logistics & Enterprise Reporting**

- Built reconciliation controls comparing source against curated totals for GL/P&L and inventory valuation, cutting discrepancies an estimated ~45% and reporting errors ~30% by resolving them at the feed (internal figures, directional rather than audited).
- Automated data-quality tests (schema, completeness, uniqueness, referential integrity) with anomaly thresholds, exception logging and pipeline alerting.
- Delivered AP and AR aging, invoice-match exceptions and credit-exposure reporting for Finance.
- Implemented FEFO and lot-level traceability for recall readiness.
- Secured self-service reporting as access widened with row-level security, RBAC and column masking.
- As sole analyst, integrated ERP, Square POS, workforce and REST API feeds into one curated data mart in Microsoft Fabric and Azure Data Factory.

### Shivam Investments
Dec 2020 – Jul 2022 · Vadodara, India (hybrid)

**Financial Analyst — Data & Reporting**

- Performed monthly GL/P&L reconciliations across sources and investigated discrepancies to keep financial reporting accurate and traceable.
- Delivered Power BI dashboards for portfolio and risk stakeholders with auditable measure definitions, and automated recurring analysis with Python and SQL, cutting manual effort an estimated ~40%.

## SELECTED PROJECTS — public code; all entities and transactions are synthetic

**Transaction Monitoring** — SQL, Python, scikit-learn, SQLite, Streamlit · synthetic data

github.com/KushPatel29/aml-transaction-monitoring · live: aml-transaction-monitoring.streamlit.app

**Layered detection over 100,299 transactions and 1,500 entities:** five explainable rules plus an anomaly model find 60 of 60 planted cases where a flat $10,000 rule finds 38, alerting on 366 entities instead of 555. The alerts become 366 controlled investigations, 78 of them P0 with a four-hour review clock, worked against five typology hypotheses that each name a lawful lookalike and the evidence that tells them apart. Scores prioritise evidence; they never decide suspicion, and the record shows zero dispositions and zero filings.

**FINTRAC mapping and an analytic brief** — PCMLTFA, LCTR, EFTR, STR · synthetic case

regulatory mapping · analytic brief INV-ENT-01216

**Threshold reports versus suspicion:** each rule mapped to FINTRAC's indicator categories and the report it relates to, including why a $10,000 threshold rule cannot see structuring. The brief works one case: six cash deposits on six consecutive days, each just under $10,000, totalling $53,374.51 in 4.9 days, nine times the subject's usual weekly cash; assessed as likely sized to avoid the threshold (moderate confidence), with the three facts needed before a reporting decision.

**GL / P&L Reconciliation** — T-SQL, SQLite, Power BI, Tableau · synthetic data

github.com/KushPatel29/gl-reconciliation-dashboard

**Controls that refuse to fake a sign-off:** a close-certification engine tests all 5,400 journal entries and enforces segregation of duties; at month-end 8 of 60 account-periods certify, 9 go to a named reviewer and 43 are blocked.

**Ask Your Data** — Python, DuckDB, Claude API · synthetic data

github.com/KushPatel29/ask-your-data · live: ask-your-data-kp.streamlit.app

**Questions over monitoring data, answered with the SQL shown:** text-to-SQL over 71 tables including an AML domain, behind a read-only guard and a default-deny access policy; on a 58-question contract it answers 46 correctly, gets none wrong and refuses 12 rather than guess.

## EDUCATION

**Northeastern University** — Master of Professional Studies in Analytics (Applied Machine Intelligence), GPA 3.76/4.0 · Vancouver, BC — **Sep 2022 – Jun 2024**

**Gujarat Technological University** — Bachelor of Engineering, Computer Science · India — **Aug 2017 – Aug 2021**

## CERTIFICATIONS

Google Data Analytics Professional Certificate · UC Davis SQL for Data Science · IBM Python for Data Science, AI & Development · Power BI Data Modelling with DAX · Databricks Lakehouse Fundamentals.